Department of Information Technology

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT
Course Code & Title: CCS334 & Big Data Analytics
Name of the Faculty member (s): Mrs.M.Rethina Kumari

Innovative Practice Description

Unit / Topic: Unit 2/ Master Slave Replication

Course Outcome: CO2

Activity Chosen: One Minute Paper

Justification:

In the one-minute paper activity, students will be given five minutes to write a brief summary of the class session. Master–Slave replication is a critical concept in Big Data that ensures data synchronization between a primary (master) database and multiple secondary (slave/replica) databases. This technique helps improve read scalability, supports failover during master crashes, and guarantees high availability. To reinforce understanding of replication architecture, synchronization mechanisms, and real-world applications, this activity is chosen.

Time Allotted for the Activity: 5 minutes
Details of the Implementation:

1. The faculty explained:
    - **Definition:** Master–Slave (Primary–Replica) Replication.
    - **Architecture:** Master database handles write operations, while one or more slaves replicate the data asynchronously or synchronously.
    - **Benefits:** Load balancing, disaster recovery, read scalability.
    - **Challenges:** Replication lag, single point of failure at the master, consistency trade-offs.
    - **Real-world Examples:** MySQL replication, MongoDB replica sets, Hadoop Distributed File System (HDFS).
2. **Student task:**
    - Each student wrote a **one-minute summary** covering at least three aspects:
      (a) How replication works,
      (b) Advantages in Big Data environments,
      (c) A limitation or challenge.
3. **Faculty review:**
    - Responses were collected and reviewed.
    - Students who confused **master–slave with peer-to-peer replication** were given feedback.
    - Corrective discussion followed to clarify **synchronous vs. asynchronous replication**.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO5	PO9	PSO1	PSO3
CO 2	3	3	2	3	1	2	2

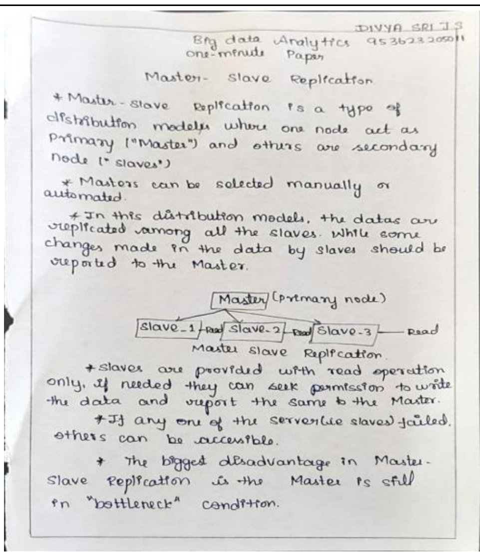
(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO5	PO9	PS01	PSO3
	3	3	2	3	1	2	2
Justification for correlation	Students will be able to recall and understand master–slave replication, which is essential for Big Data system design.	Students will apply the learned concepts in database modeling and replication strategies.	Students will develop solutions for data reliability and fault tolerance using replication techniques	Individual involvement improves critical thinking and self-assessment.	Emphasize continuous learning by keeping up with advancements in distributed database tools (e.g., Hadoop, MongoDB, Cassandra).	Develop precise and accurate data replication strategies for scalability.	Ensure compliance with performance and availability standards in Big Data systems.

- **Images / Screenshot of the practice:**

Figure 1 & 2: shows the Students Presenting their understanding in the paper



DIVYA SRI J S
Big data Analytics 953623205011
one-minute Paper

Master- Slave Replication

* Master - Slave Replication is a type of distribution models where one node act as Primary ("Master") and others are secondary node ("slaves")

* Masters can be selected manually or automated.

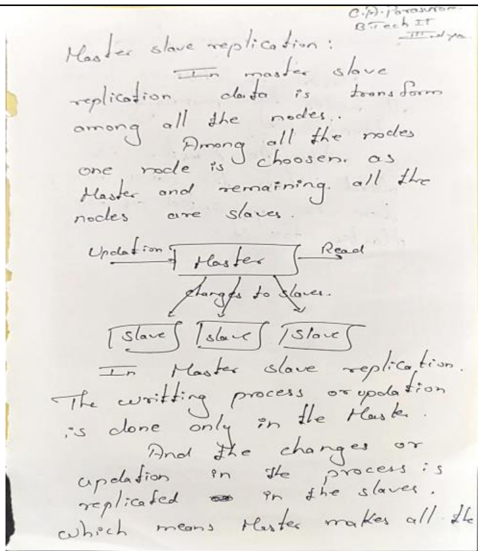
* In this distribution models, the datas are replicated among all the slaves. While some changes made in the data by slaves should be reported to the Master.

Master (primary node)
Slave_1 Read Slave-2 Read Slave-3 Read
Master slave Replication.

* slaves are provided with read operation only, if needed they can seek permission to write the data and report the same to the Master.

* If any one of the server(ie slaves) failed, others can be accessible.

* The biggest disadvantage in Master-Slave Replication is the Master is still in "bottleneck" condition.



C.A. Parasuram
B.Tech IT
III yr

Master slave replication:
In master slave replication data is transform among all the nodes.
Among all the nodes one node is choosen as Master and remaining all the nodes are slaves.

Updation Master Read
changes to slaves.
Slave Slave Slave

In Master slave replication. The writting process or updation is done only in the Master.
And the changes or updation in the process is replicated in the slaves. which means Master makes all the

Figure 3 & 4: shows the proof of the students Dhivya Sri.J and Parasuram C.A.

❖ **Feedback of practice from students and other stakeholders:**

- Students reported that summarizing **replication concepts** helped reinforce their understanding of Big Data system architecture.
- Students felt the activity was beneficial for improving recall and would help them in future **database and distributed system courses**.

❖ **Benefit of the practice:**

- Improved retention and understanding of **master–slave replication in Big Data**.
- Identification of knowledge gaps in replication and consistency models, which will be addressed in future sessions.

❖ **Challenges faced in implementation:**

- Some students initially struggled with summarizing technical replication concepts concisely.
- A few students required additional time to complete the activity.

References:

1. https://www.geeksforgeeks.org/system-design/types-of-database-replication-system-design/?utm_source=chatgpt.com
2. https://dennylesmana.medium.com/master-slave-replication-database-concept-for-beginners-300b3f9a8228
3. https://www.enjoyalgorithms.com/blog/master-slave-replication-databases?utm_source=chatgpt.com

Signature of Faculty Member **HOD**